

Amjad Althabteh

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Summary

C++ engineer focused on real-time graphics, compiler tooling, and performance-critical systems. Engineered rendering engines, simulation systems, and diagnostics tools using C++17–C++23, OpenGL, and modern systems design. Driven by low-level optimization, engine architecture, and solving complex problems close to the hardware.

Education

B.S. in Computer Science | Wayne State University
Detroit, MI | GPA: 3.65/4.00

Experience

Graphics Engineer	Immersalab Graphics	Detroit, MI	Sep 2025 - Dec 2025
- Optimized GPU resource handling and frame pipeline, improving rendering stability - Designed internal C++/JSON tooling and asset pipeline; profiled GLSL shader compilation paths - Reduced GPU idle time by ~30% via data-driven pipeline optimizations across test scenes - Engineered GPU profiling and frame-timing tools, accelerating bottleneck diagnosis			
Unreal C++	Independent Game	Ann Arbor, MI	Apr 2025 - Aug 2025
- Engineered gameplay systems in C++ and Unreal Engine for combat, movement, and interactions - Designed animation state machines for multiple characters, improving transition responsiveness - Optimized animation systems enabling smooth multi-layer blending across 15+ states			
Software Development Intern	Rare Munchies	Ann Arbor, MI	Sep 2023 - Apr 2024
- Integrated TypeScript UI components, improving page transitions and responsiveness - Identified and resolved 8+ cross-browser compatibility issues. - Optimized client-side rendering and state management, reducing re-renders by ~35%			

Undergraduate Researcher | Orbital Collision Modeling
Wayne State University | Detroit, MI | Feb 2025 – Dec 2025

- Modeled 50+ satellite trajectories using numerical integration and N-body dynamics
- Simulated 1M+ orbital trajectories using spatial partitioning, reducing compute time by ~40%

Projects

Glint 3D | Rendering Engine [glint-rendering](#)

- Reduced heap allocations via custom memory pool (placement new + freelist)
- Engineered a C++17/OpenGL rendering engine with scene graph + pipeline
- Implemented frustum culling and instanced rendering (glDrawElementsInstanced)

3D Space Simulation Engine [orbital_collision_engine](#)

- Engineered a modular C++17/OpenGL simulation engine for orbital systems
- Designed rendering, physics, math, and entity systems with clean separation of concerns
- Built custom math library (Vec3, Mat4) and camera system for 3D transformations

HFT OrderBook Engine

- Built a FIFO order matching engine in C++20 for bid/ask execution workflows
- Used alignas(64) cache-aligned bid/ask structs to eliminate false sharing and reduce memory latency.

Traceforge | Runtime Stack Trace & Diagnostics Engine

- Engineered a C++23 diagnostics engine using std::stacktrace to improve runtime debugging visibility.
- Implemented custom memory tracking (new/delete) for leak detection
- Automated regression snapshot diffing using std::filesystem across build artifacts.

Skills

Languages: C++ (C++11–C++23), Rust, C, C#, Python, TypeScript, Java, SQL

Graphics & Engines: OpenGL, GLSL, Unreal Engine, Unity, Godot

Systems: Custom allocators, memory pools, multithreading, lock-free data structures, cache optimization, profiling